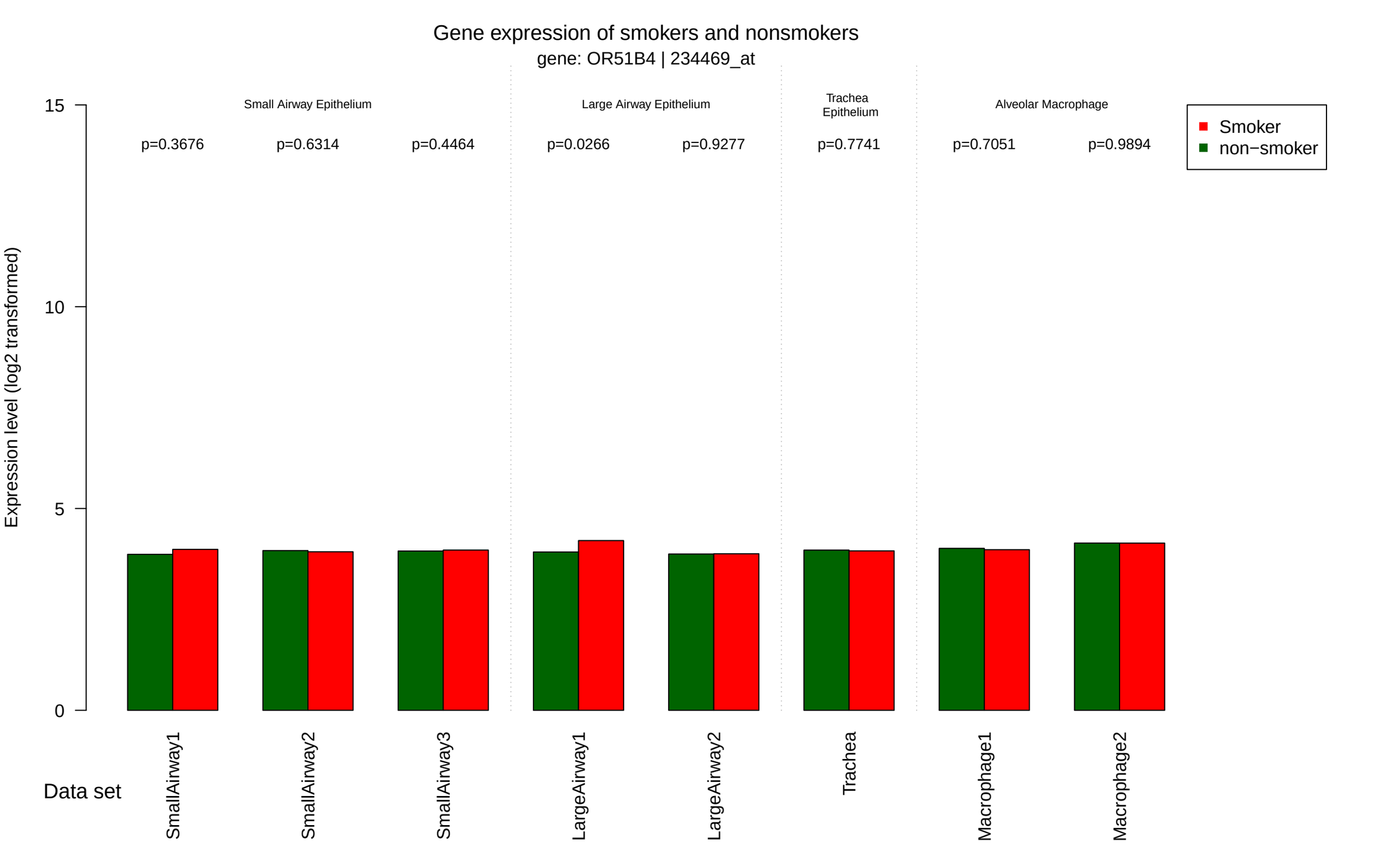
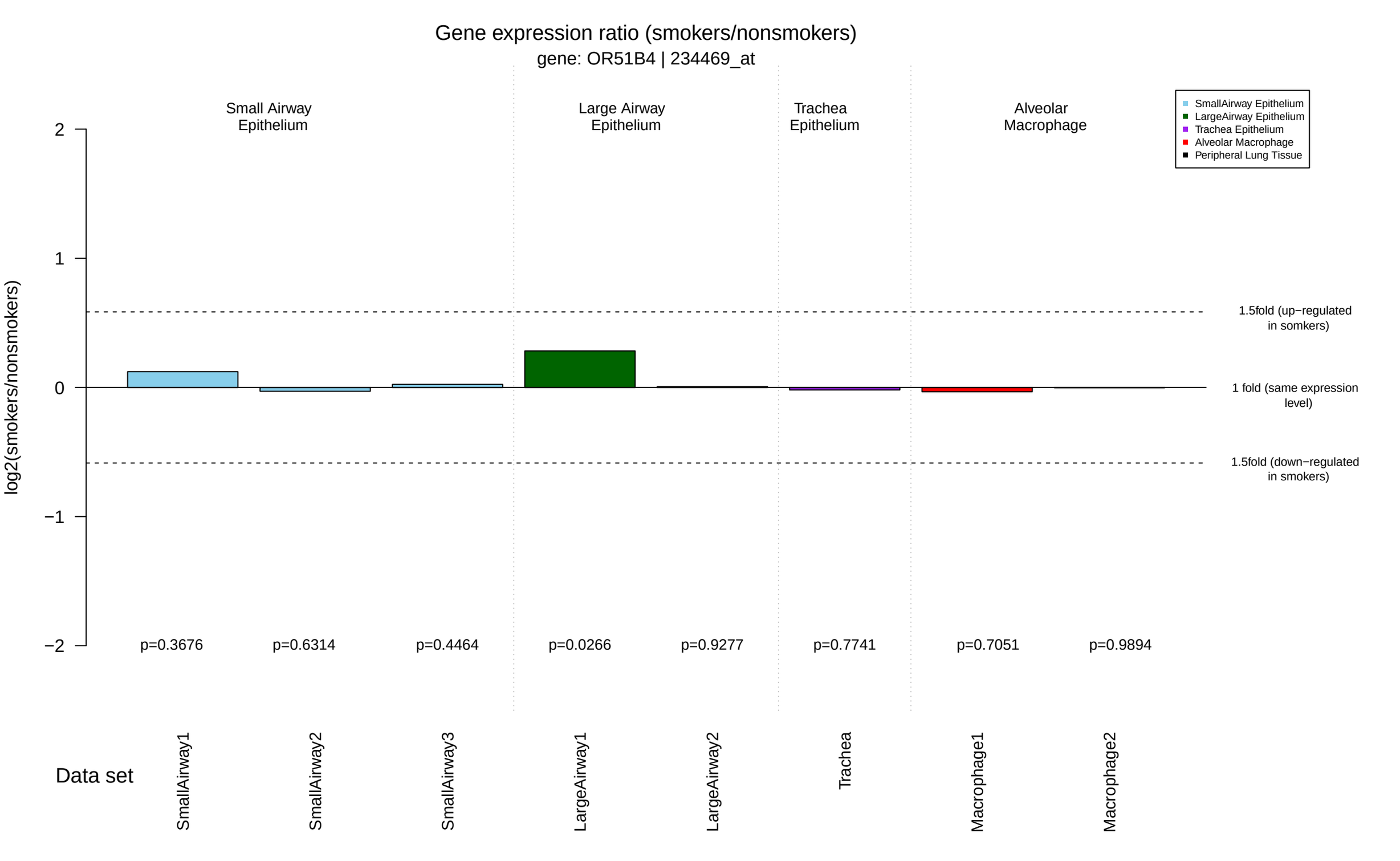
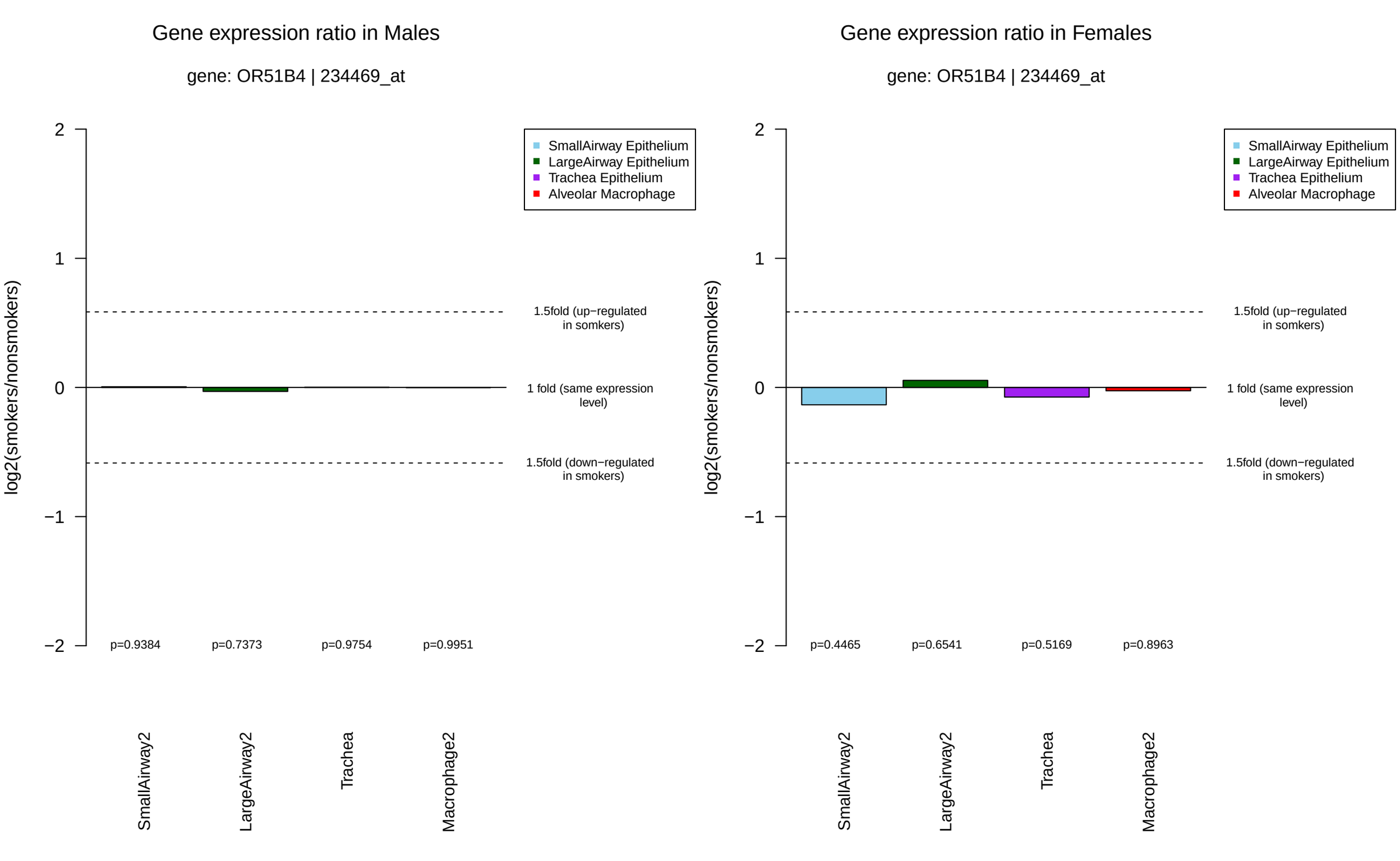
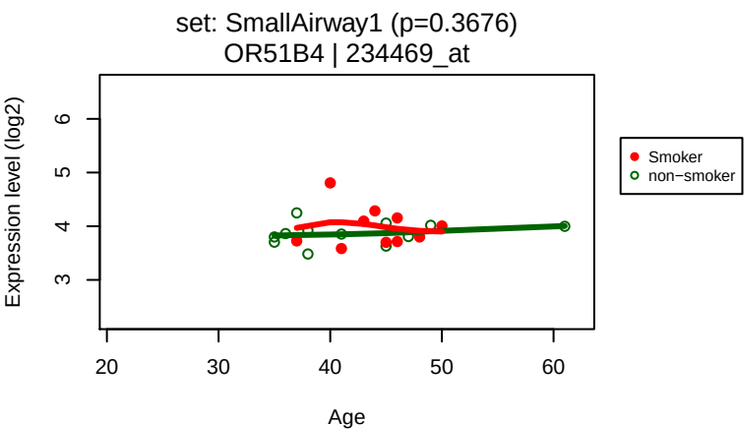
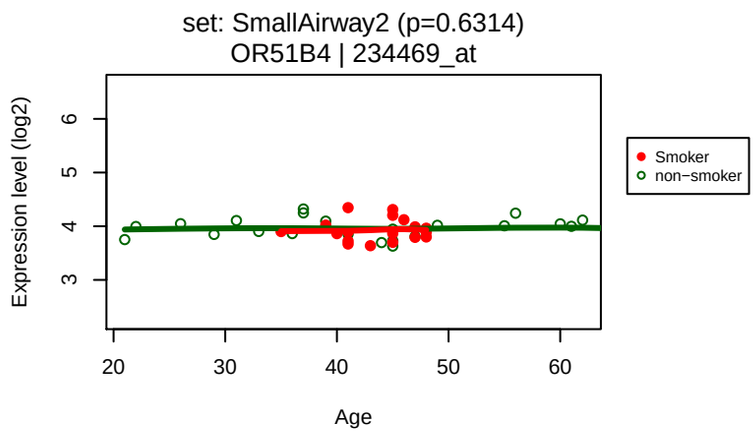
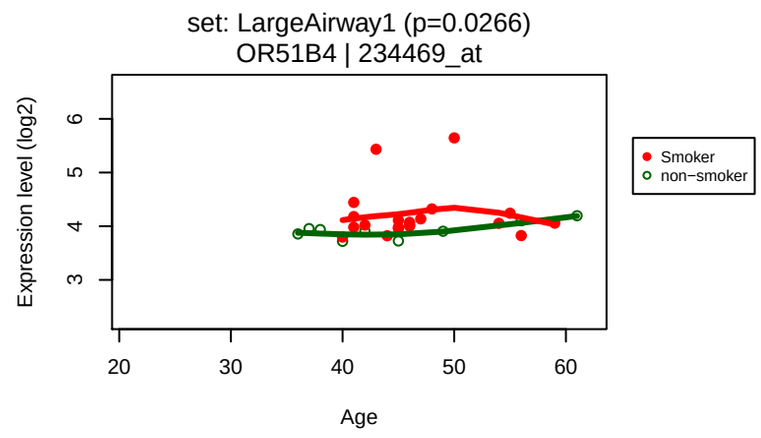

Gene expression of smokers and nonsmokers
gene: OR51B4 | 234469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3676
p=0.6314
p=0.4464
p=0.0266
p=0.9277
p=0.7741
p=0.7051
p=0.9894
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

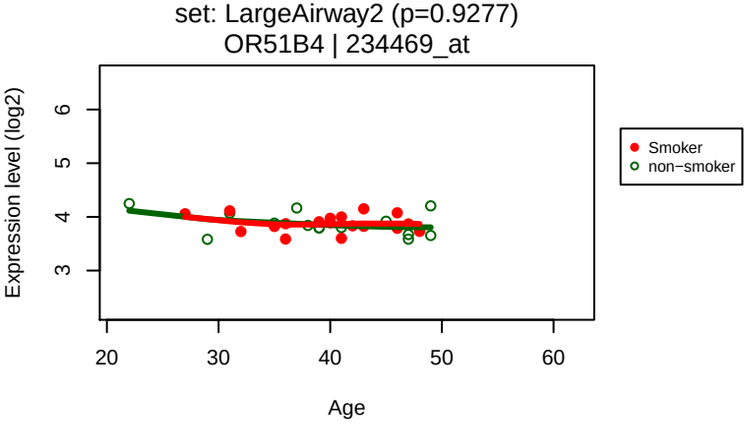

Gene expression ratio (smokers/nonsmokers)
gene: OR51B4 | 234469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3676
p=0.6314
p=0.4464
p=0.0266
p=0.9277
p=0.7741
p=0.7051
p=0.9894
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

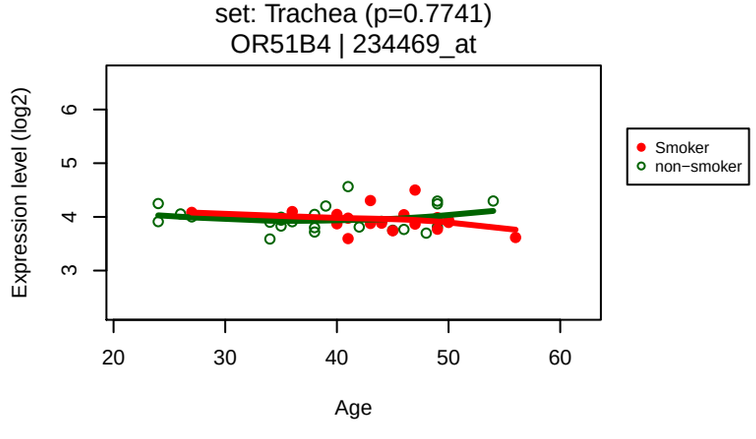

Gene expression ratio in Males
gene: OR51B4 | 234469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9384
p=0.7373
p=0.9754
p=0.9951
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51B4 | 234469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4465
p=0.6541
p=0.5169
p=0.8963
SmallAirway2
LargeAirway2
Trachea
Macrophage2

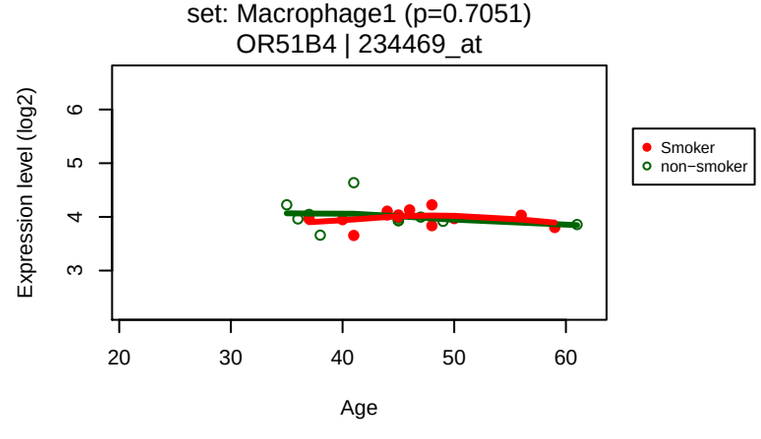

set: SmallAirway1 (p=0.3676)
OR51B4 | 234469_at
Expression level (log2)
Age
Smoker
non−smoker

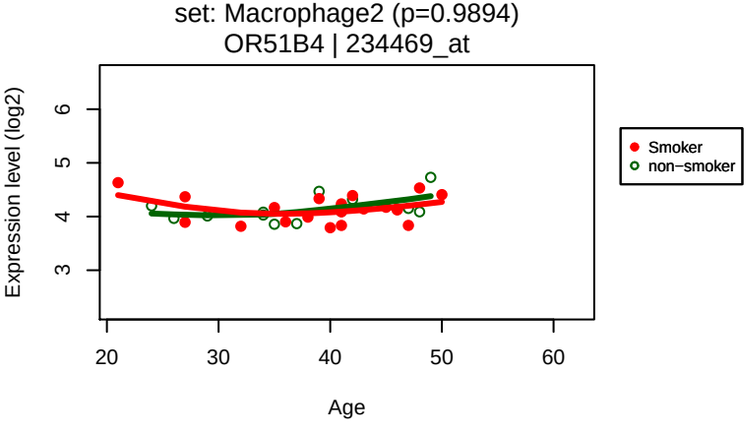

set: SmallAirway2 (p=0.6314)
OR51B4 | 234469_at
Expression level (log2)
Age
Smoker
non−smoker

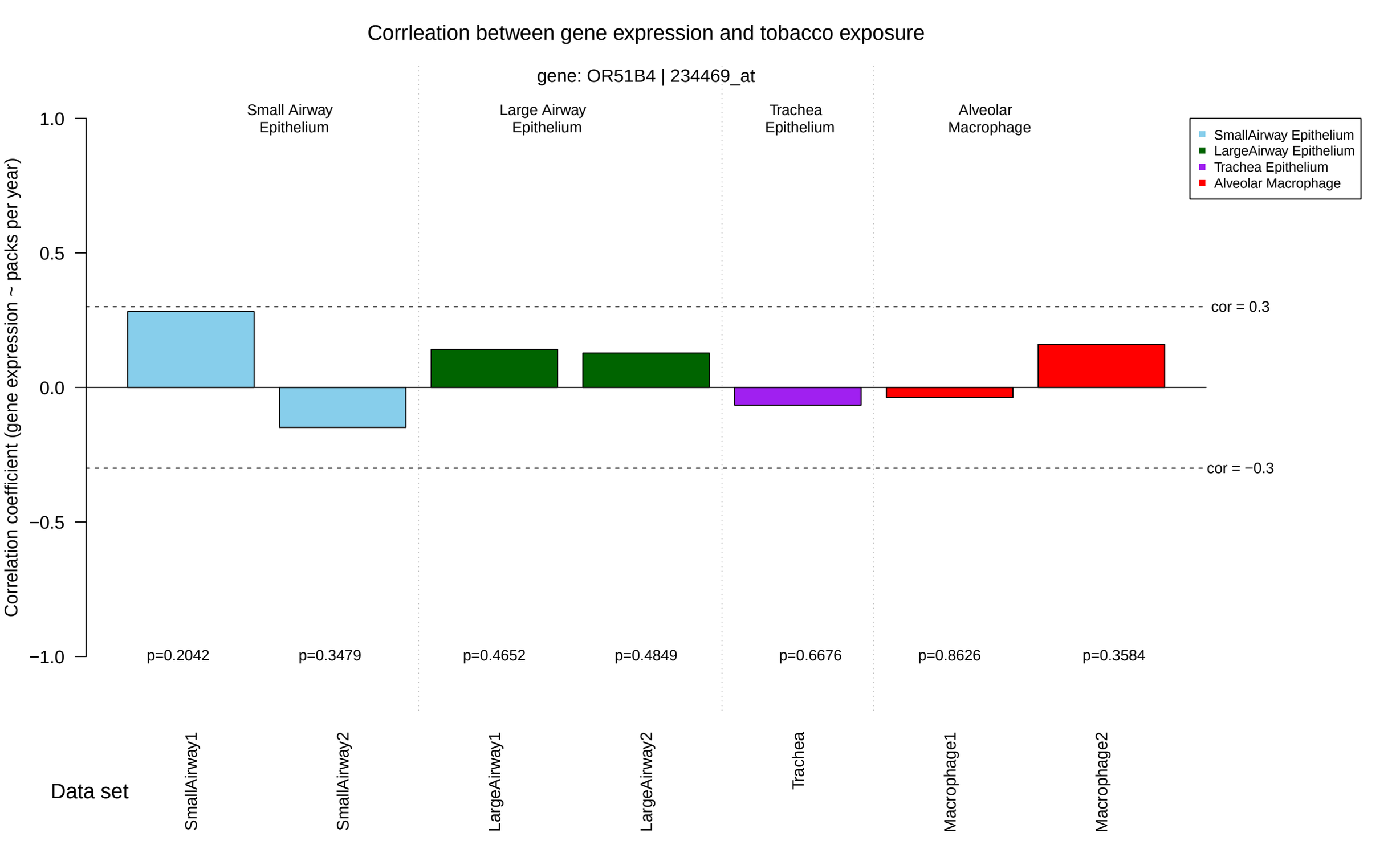

set: LargeAirway1 (p=0.0266)
OR51B4 | 234469_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9277)
OR51B4 | 234469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7741)
OR51B4 | 234469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7051)
OR51B4 | 234469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9894)
OR51B4 | 234469_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51B4 | 234469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2042
p=0.3479
p=0.4652
p=0.4849
p=0.6676
p=0.8626
p=0.3584
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2